

K2-29b

R-2085

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_221106 · created 2026-08-10T00:28:38+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459889.8342 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.013
Transit depth (Rp/Rs)^2	1.63 +/- 0.33 [%]
Orbital Inclination (inc)	87.09 +/- 1.03
Airmass coefficient 1 (a1)	0.836 +/- 0.0073
Airmass coefficient 2 (a2)	0.173 +/- 0.089
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	#2 - [517, 359]
Optimal Method	PSF photometry
Transit Duration (day)	0.0825 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.128 ± 0.013 vs 0.1419 (deviation 1.07σ)
Duration fit vs published	0.0825 d vs 0.0925 d (deviation 1.54σ)
Mid-time O-C	-9.4 min
Depth SNR	9.8
Residual scatter	0.88 %

Light curve

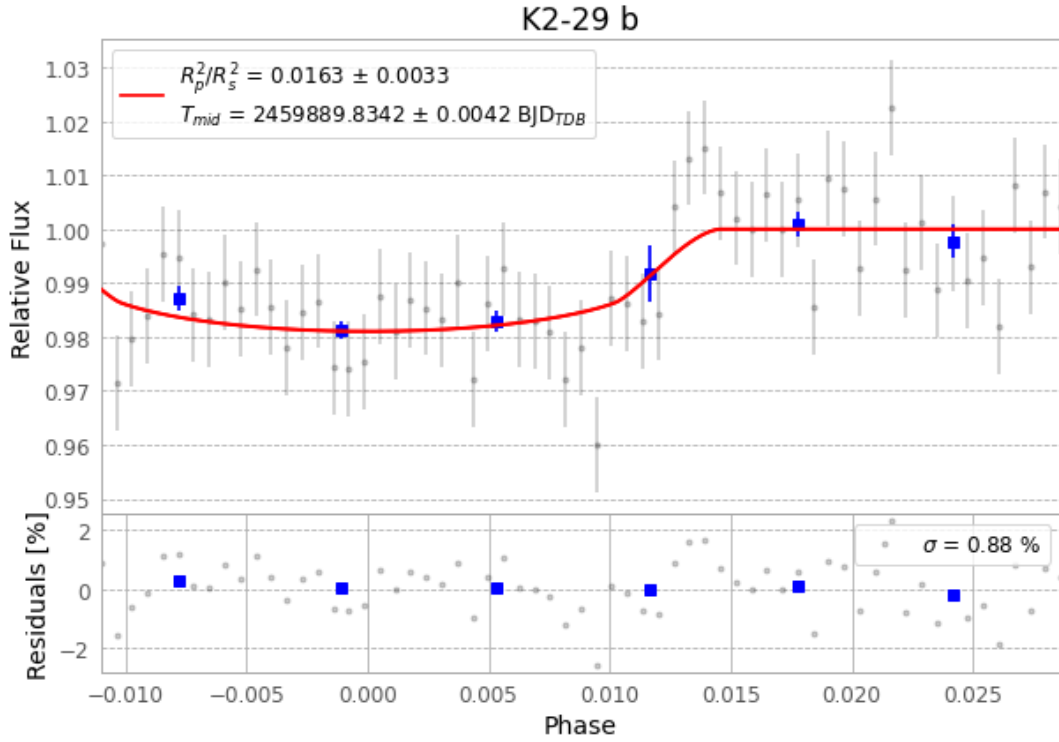


Figure 1 — FinalLightCurve_K2-29 b_2022-11-06.png (SHA-256 5f11003aaea5f7f9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 162], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 55631f412fb2559b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

63 FITS frames (41 MB), K2-29221106070017.FITS → K2-29221106100615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-221106.log (4f65acfcbe86...), FinalLightCurve_K2-29 b_2022-11-06.png (5f11003aaea5...), FinalLightCurve_K2-29 b_2022-11-06.csv (f812363e5d5b...)

Compute resources

Wall time 208.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:28Z.